

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	
Project Name	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	Proper indentation
2	Proper File Structure	Files and folders are logically organized	Yes	Well organized
3	Meaningful Variable Name	Variables reflect their purpose clearly	Yes	Meaningful names
4	Function / Method Names	Functions are descriptively named	Yes	Clear functions
5	Code Comments	Inline and block comments explain logic	Yes	Comments added
6	Modular Design	Code is split into reusable functions/modules	Yes	Modular design
7	No Redundant Code	Duplicate or unused code is removed	Yes	No duplicate code
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Proper routing

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
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1	Flask Routes	Python, Flask	Home, About, Prediction pages	High
2	Machine Learning	Scikit-Learn	Crop Prediction	High
3	HTML Templates	HTML	All webpages	High
4	CSS Styling	CSS	Entire website	High
5	Prediction Function	Python	Prediction module	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate folders for templates, static files, dataset, model, and
Readability	5	Meaningful variable names, proper indentation, and comments improve readability.
Reusability	5	Reusable Flask routes, CSS, and prediction logic are used throughout the project.
Documentation / Comments	5	The code includes sufficient comments and follows a modular structure.
Overall Score	5/5	The project is well-structured, readable, reusable, and easy to maintain.